

МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ

НАЦІОНАЛЬНИЙ ТЕХНІЧНИЙ УНІВЕРСИТЕТ
«ХАРКІВСЬКИЙ ПОЛІТЕХНІЧНИЙ ІНСТИТУТ»

Звіт

з лабораторної роботи № 7

з дисципліни «Стек технологій проєктування WEB-застосунків»

Виконав:

студент групи ІКМ-М225В

Суліма Нікіта Володимирович

Перевірив:

аспірант каф. КМПС Хорошун Андрій Сергійович

Харків 2026

Console

Commands

CreateTaskInteractive

```
<?php

namespace App\Console\Commands;

use Illuminate\Console\Command;
use App\Models\Task;

class CreateTaskInteractive extends Command
{
    protected $signature = 'tasks:create-interactive';

    protected $description = 'Create task interactively';

    public function handle()
    {
        $title = $this->ask('Введіть назву задачі');
        $description = $this->ask('Короткий опис (необов\'язково)');
        $dueDate = $this->ask('Дата дедлайну (YYYY-MM-DD)');
        $status = $this->choice('Оберіть статус', ['new', 'in_progress', 'done'], 0);
        $assignee = $this->ask('ID виконавця (або залиште порожнім)');

        if (!$this->confirm('Створити цю задачу?', true)) {
            $this->warn('Скасовано');
            return;
        }

        $task = Task::create([
            'title' => $title,
```

```

        'description' => $description,
        'status' => $status,
        'project_id' => 1, // можна змінити або запитати окремо
        'author_id' => 1, // можна прив'язати до auth
        'assigned_to' => $assignee ?: null,
        'due_date' => $dueDate,
    ];

    $this->info("Задача '{$task->title}' створена з ID: {$task->id}");
}
}

```

GenerateTaskReport

```

<?php

namespace App\Console\Commands;

use Illuminate\Console\Command;
use App\Models\Task;

class GenerateTaskReport extends Command
{
    protected $signature = 'tasks:report [--project_id=]';

    protected $description = 'Generate tasks report';

    public function handle()
    {
        $projectId = $this->option('project_id');

        $query = Task::query();

        if ($projectId) {
            $query->where('project_id', $projectId);
        }
    }
}

```

```

    }

    $tasks = $query->get();

    if ($tasks->isEmpty()) {
        $this->warn('Задач не знайдено');
        return;
    }

    $this->info('Список задач');

    $this->table(
        ['ID', 'Назва', 'Статус', 'Дедлайн'],
        $tasks->map(function ($task) {
            return [
                $task->id,
                $task->title,
                $task->status,
                $task->due_date,
            ];
        })
    );
}
}

```

Migrations

0001_01_01_000000_create_users_table

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration

```

```
{
  /**
   * Run the migrations.
   */
  public function up(): void
  {
    Schema::create('users', function (Blueprint $table) {
      $table->id();
      $table->string('name');
      $table->string('email')->unique();
      $table->timestamp('email_verified_at')->nullable();
      $table->string('password');
      $table->rememberToken();
      $table->timestamps();
    });

    Schema::create('password_reset_tokens', function (Blueprint
$table) {
      $table->string('email')->primary();
      $table->string('token');
      $table->timestamp('created_at')->nullable();
    });

    Schema::create('sessions', function (Blueprint $table) {
      $table->string('id')->primary();
      $table->foreignId('user_id')->nullable()->index();
      $table->string('ip_address', 45)->nullable();
      $table->text('user_agent')->nullable();
      $table->longText('payload');
      $table->integer('last_activity')->index();
    });
  }
}
```

```
/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::dropIfExists('users');
    Schema::dropIfExists('password_reset_tokens');
    Schema::dropIfExists('sessions');
}
};
```

0001_01_01_000001_create_cache_table

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('cache', function (Blueprint $table) {
            $table->string('key')->primary();
            $table->mediumText('value');
            $table->integer('expiration');
        });

        Schema::create('cache_locks', function (Blueprint $table) {
            $table->string('key')->primary();
            $table->string('owner');
```

```

        $table->integer('expiration');
    });
}

/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::dropIfExists('cache');
    Schema::dropIfExists('cache_locks');
}
};

```

0001_01_01_000002_create_jobs_table

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('jobs', function (Blueprint $table) {
            $table->id();
            $table->string('queue')->index();
            $table->longText('payload');
            $table->unsignedTinyInteger('attempts');
            $table->unsignedInteger('reserved_at')->nullable();

```

```
    $table->unsignedInteger('available_at');
    $table->unsignedInteger('created_at');
});
```

```
Schema::create('job_batches', function (Blueprint $table) {
    $table->string('id')->primary();
    $table->string('name');
    $table->integer('total_jobs');
    $table->integer('pending_jobs');
    $table->integer('failed_jobs');
    $table->longText('failed_job_ids');
    $table->mediumText('options')->nullable();
    $table->integer('cancelled_at')->nullable();
    $table->integer('created_at');
    $table->integer('finished_at')->nullable();
});
```

```
Schema::create('failed_jobs', function (Blueprint $table) {
    $table->id();
    $table->string('uuid')->unique();
    $table->text('connection');
    $table->text('queue');
    $table->longText('payload');
    $table->longText('exception');
    $table->timestamp('failed_at')->useCurrent();
});
}
```

```
/**
```

```
 * Reverse the migrations.
```

```
 */
```

```
public function down(): void
{
```



```
Schema::dropIfExists('jobs');
Schema::dropIfExists('job_batches');
Schema::dropIfExists('failed_jobs');
}
};
```

2026_05_03_111849_create_projects_table

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('projects', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->text('description')->nullable();
            $table->foreignId('user_id')->constrained()-
>cascadeOnDelete();
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
```

```
{
    Schema::dropIfExists('projects');
}
};
```

2026_05_03_112137_create_tasks_table

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('tasks', function (Blueprint $table) {
            $table->id();
            $table->string('title');
            $table->text('description')->nullable();
            $table->string('status')->default('new');

            $table->foreignId('project_id')
                ->constrained()
                ->cascadeOnDelete();

            $table->foreignId('author_id')
                ->constrained('users');

            $table->foreignId('assigned_to')
                ->nullable()
        });
    }
};
```

```

        ->constrained('users');

        $table->date('due_date')->nullable();

        $table->timestamps();
    });
}

/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::dropIfExists('tasks');
}
};

```

2026_05_03_112554_create_comments_table

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('comments', function (Blueprint $table) {
            $table->id();

```

```

        $table->foreignId('task_id')
            ->constrained()
            ->cascadeOnDelete();

        $table->foreignId('user_id')
            ->constrained()
            ->cascadeOnDelete();

        $table->text('content');

        $table->timestamps();
    });
}

/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::dropIfExists('comments');
}
};

```

2026_05_03_112609_create_files_table

```

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.

```

```
*/  
public function up(): void  
{  
    Schema::create('files', function (Blueprint $table) {  
        $table->id();  
  
        $table->string('entity'); // наприклад: 'task', 'project'  
        $table->unsignedBigInteger('entity_id');  
  
        $table->string('file_path');  
        $table->string('original_name');  
  
        $table->timestamps();  
  
        // опційно (але корисно для швидкості пошуку)  
        $table->index(['entity', 'entity_id']);  
    });  
}  
  
/**  
 * Reverse the migrations.  
 */  
public function down(): void  
{  
    Schema::dropIfExists('files');  
}  
};
```